# KRONE: Hierarchical and Modular Log Anomaly Detection

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SECTION 1

# INTRODUCTION

# Why Log Anomaly Detection Matters?

*Logs are software's nervous system — unreadable at scale without automation.*

### Security & Reliability

Ground truth for intrusions and SLA breaches.

### Scale & Complexity

TB-scale daily across hundreds of services.

### Operational Cost

Outages cost thousands/minute; alerts drown real incidents.

# The Problem of Flat Log AD Paradigm

Log sequence

| | | | | | | |
|---|---|---|---|---|---|---|
| $L_1$ | $L_2$ | $L_3$ | $L_4$ | $L_5$ | $L_6$ | … |

**billions per hour !**

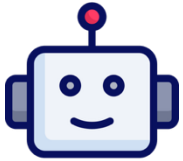
**Deep Learning Models**

LSTM[1], Transformer[2]…

Anomaly score = 0.6

**Uninterpretable Detection**

**LLM detection [3,4]**

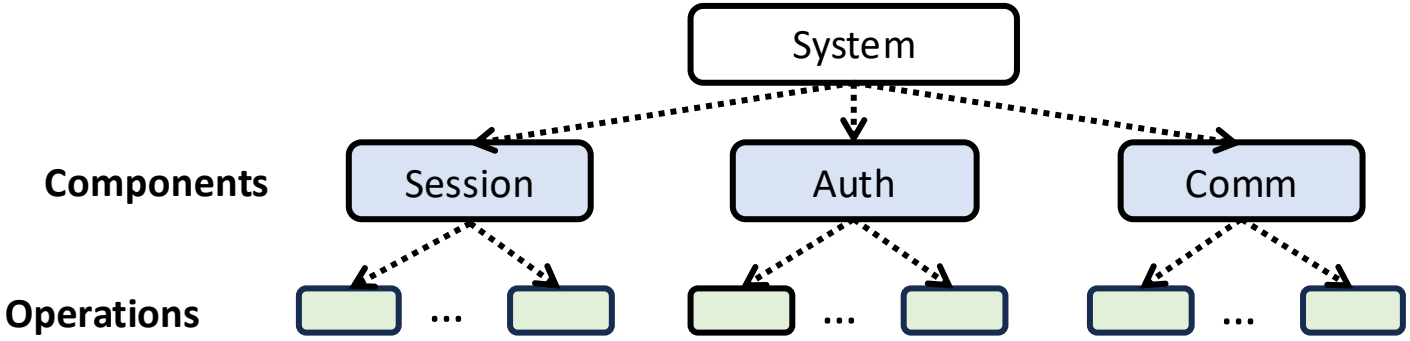


Context

**Context window limit & Huge Cost!**

[1 -4] See Reference Page.

# However, System Executions Are Hierarchical



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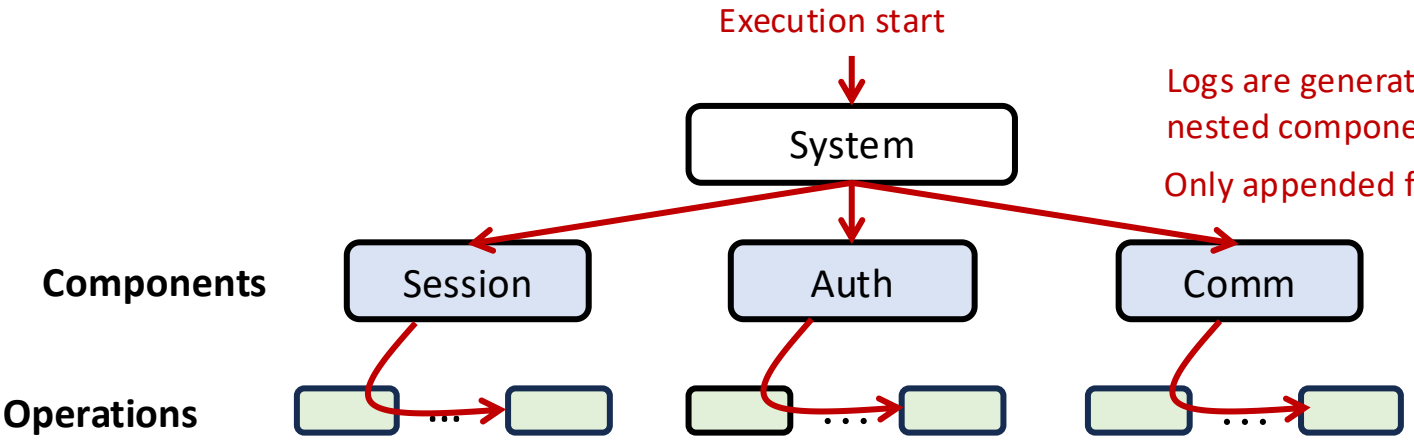



Logs are generated **hierarchically** by nested components and operations.

Only appended flatly.

**Logs**

| | | |
|---|---|---|
| 1 | Open session started | Session |
| 2 | Session Opened successfully | |
| 3 | Auth start | Auth |
| 4 | Auth succeeds | |
| 5 | Comm: GET_request | Comm |
| 6 | Comm: GET_response | |

Logs for session operations

Logs for Auth operations

Logs for Comm operations

# An Example of Hierarchical Anomaly Detection

**Hierarchy = natural execution boundaries for structured anomaly detection**

- **Low: Intra-Component**

| | |
|---|---|
| 1 | Open session started |
| 2 | Session Opened successfully |
| 3 | Auth start |
| 5 | Comm: GET_request |
| 6 | Comm: GET_response |

Session ✓

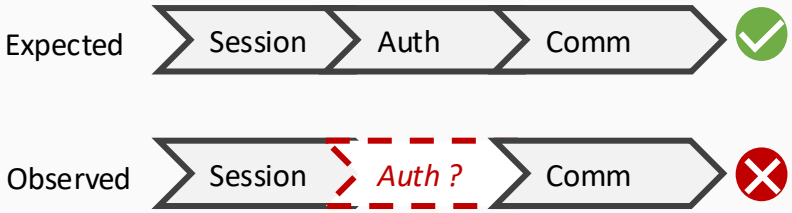
Auth ✗

Comm ✓

**Intra-component Anomaly**

*Sessions look fine, but Auth is incomplete*

- **High: Across-Component**




**Inter-component Anomaly**

*Each component runs fine, but Auth never replies*

# Benefits of the Hierarchical Log AD Paradigm

## 1 Modular Detection

- Reusable atomic units + permutations
- Sequence detection → modular local tasks

## 2 Scalable LLM Log AD

- LLMs run on compact semantic-coherent units, not full traces
- Lower cost, latency, context

## 3 Localization & Explanation

- Anomalies bound to execution scope
- Grounded in semantic meaning
- Clearer root-cause analysis



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# KRONE Leads on Accuracy, Resource-Efficiency, and Interpretability

Evaluated on three public benchmarks and an industrial dataset from ByteDance Cloud — comprehensive gains across four dimensions:

**Accuracy**

**42.49% → 87.98%**

F1 of same detector w/ vs w/o hierarchy

**Data Efficiency**

**117.3× ↓**

reduction in data space

**LLM Resource Efficiency**

**55.7 × ↓**

lower LLM cost compared to exhaustive LLM detection method [5]

**Interpretability**

**✓**

clearer anomaly localization & explanation

**F1-score: 82.76% → 92.83% (+10.07% over prior methods)**

LLM usage reduced to just 1.1%–3.3% of test data size

[5] See Reference Page.

# Challenges of Hierarchical Log Anomaly Detection

## C1 Abstracting Hierarchy from Black-Box Systems

*No structure, no domain knowledge*

- Systems expose no explicit structure
- Infer hierarchy from logs alone

## C2 Lightweight Modeling 4 Massive Logs

*Billions of logs, no hard-coded rules*

- Heavy models infeasible at scale
- Manual rule maintenance breaks down
- Need efficient, adaptive abstraction

## C3 Detection under Nested Semantics

*Same high-level, different low-level*

- High-level only: fast but coarse
- Incorporating Low-level detail: accurate but costly
- Balance fidelity with efficiency



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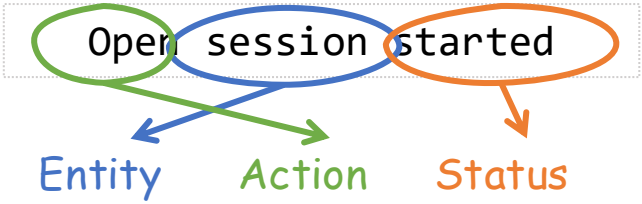
SECTION 2

# KRONE LOG ABSTRACTION MODEL

# Our Observation — Hidden Hierarchy in Log Semantics

An example log message




A log message usually describes a **status** as the outcome of an **action** on an **entity.**

**Log Templates**

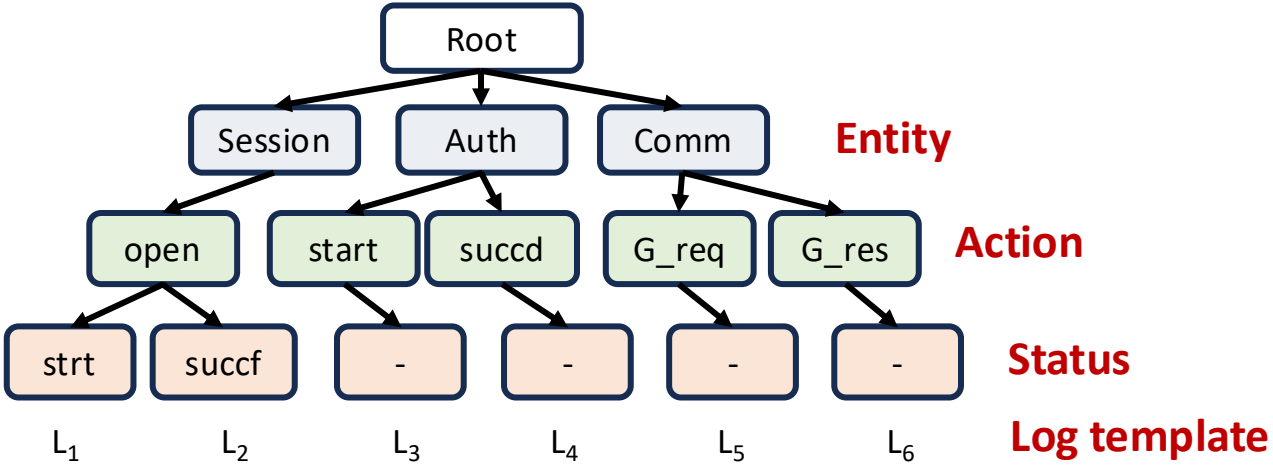
| Index | Log Template |
|---|---|
| 1 | Open session started |
| 2 | Session Opened successfully |
| 3 | Auth start |
| 4 | Auth succeeds |
| 5 | Comm: GET_request |
| 6 | Comm: GET_response |

**Semantic Structure**

| Entity | Action | Status |
|---|---|---|
| Session | Open | started |
| | | succfl |
| Auth | start | - |
| | succeed | - |
| Comm | GET_req | - |
| | GET_res | - |

# KRONE Log Abstraction Model

**KRONE-Tree:** Encodes the relationships between entity, action, status, and the log templates.

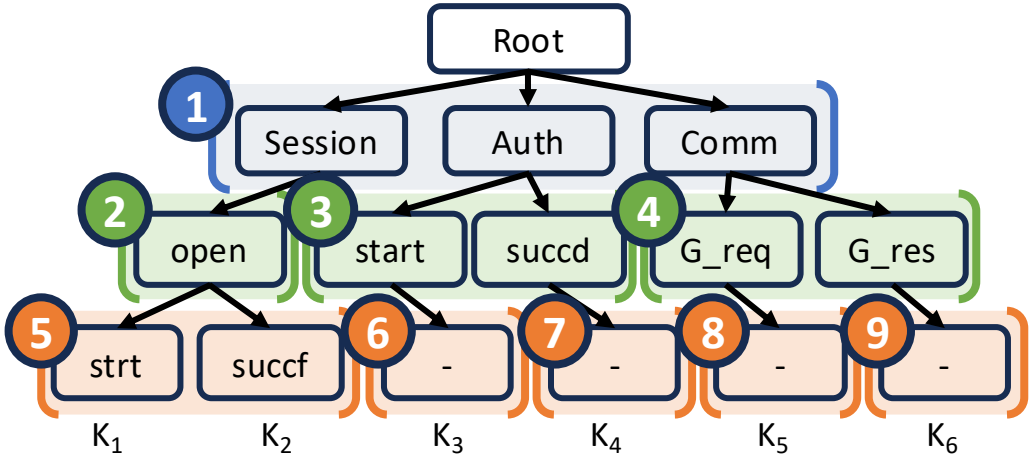


# KRONE Log Abstraction Model

**Log Sequence Decomposition**: Log sequence → structured set of KRONE-Seqs

**Log Sequence Detection → Modular detections on KRONE-Seqs**

Seq = [$L_1$, $L_2$, $L_3$, $L_4$, $L_5$, $L_6$]




**KRONE-Seqs**

1. [Session, Auth, Comm]
2. Session→[open]
3. Auth → [start, succeeds]
4. Comm → [GET_request, GET_response]
5. Session → open →[started, successful]
6. Auth → start → [-]
7. Auth → succeeds → [-]
8. Comm → GET_request →[-]
9. Comm → GET_response →[-]

x Krone E-Seq  x Krone A-Seq  x Krone S-Seq

SECTION 3

# KRONE FRAMEWORK

# What is KRONE?

*A foundational **orchestration framework** for log anomaly detection that **decomposes, executes, optimizes, and aggregates** modular sub-problems*

**Decompose**

Split a log-anomaly task into modular sub-problems

**Execute**

Run each modular sub-problem through detector operators

**Optimize**

Optimize sub-task execution via **knowledge reuse** and **early stopping**

**Aggregate**

Combine operator outputs into a final prediction

**Detectors-as-Operators:** any detector—pattern-based, deep learning, or LLM-based—can be seamlessly plugged into KRONE as a **modular operator**.



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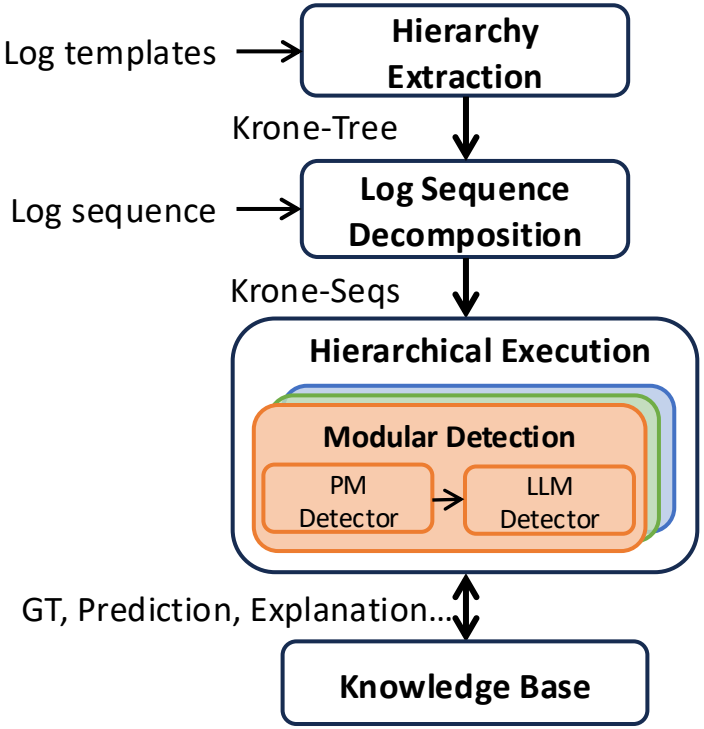
# KRONE Framework Overview




Extract entity, action, status per log template

Decompose log sequence into krone-seqs

Decompose

Hybrid Modular detection for Krone-seqs

Hierarchical execution of the modular detections

Execute, optimize, aggregate

Knowledge caching and reusing for LLM resource optimization

Cache & Reuse

# Hierarchy Extraction

| **Hierarchy Extraction** | Log Sequence Decomposition | Modular Detection | Hierarchical Exec & Opt | Knowledge Base |
|---|---|---|---|---|

- **Goal:** Turn each log template into a structured representation
- **How:** LLM extracts **(entity, action, status)** triples per template
- **Adaptive and One-time cost:** Done once per log template as pre-processing

# Log Sequence Decomposition

| Hierarchy Extraction | **Log Sequence Decomposition** | Modular Detection | Hierarchical Exec & Opt | Knowledge Base |
|---|---|---|---|---|

- **Goal:** Split a long log sequence into execution units (KRONE-Seqs)
- **Output:** A set of multi-level **KRONE-Seqs** (status-, action-, entity-)
- **Why:** Localizes anomaly detection — each KRONE-Seq can be evaluated independently, improving precision and enabling caching



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# Modular Detection

| Hierarchy Extraction | Log Sequence Decomposition | **Modular Detection** | Hierarchical Exec & Opt | Knowledge Base |
|---|---|---|---|---|

- **Two-stage detection** per **KRONE-Seq** — fast filter, then deep check
  - **PM detector:** pattern-match against known normal KRONE-Seqs decomposed from the training set (stored in KB), filters out the easy cases instantly, no LLM call
  - **LLM detector:** for the remaining unknown KRONE-Seqs, in-context learning using KB normals as demonstrations
- **Benefit:** Most real-world KRONE-Seqs are **recurring** — PM handles them cheaply; LLM only engages where necessary.



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# Hierarchical Execution & Optimization

| Hierarchy Extraction | Log Sequence Decomposition | Modular Detection | **Hierarchical Exec & Opt** | Knowledge Base |
|---|---|---|---|---|

- **Bottom-up execution:** Detect at **status → action → entity**, lowest level first

  Greedily minimize cost

- **Early stop:** stop level up when anomaly detected

  Avoid unnecessary computation costs

- **Result:** Substantially fewer LLM calls at inference time — hierarchy pays off in efficiency

# Knowledge Base

| Hierarchy Extraction | Log Sequence Decomposition | Modular Detection | Hierarchical Exec & Opt | **Knowledge Base** |
|---|---|---|---|---|

- **Built from training data:** Decomposed KRONE-seqs from normal training data serve as **GT** and in-context **demonstrations** for the LLM detector
- **Grows at test time:** store LLM predictions on unseen test KRONE-seqs
- **Amortization effect:** prediction retrieval for recurring test KRONE-seqs — LLM cost amortizes over inference time!



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SECTION 4

# EVALUATION

# Datasets and Highlight Results

Evaluated on **HDFS, BGL, Thunderbird (public)** and **ByteDance-IaaS (industrial)** — KRONE delivers across four dimensions:

| Reusability | Data Efficiency | Accuracy | LLM Efficiency |
|---|---|---|---|
| **876.4×** | **117.3× ↓** | **82.76% → 92.83%** | **55.07× ↓** |
| total occurrences vs. unique KRONE-Seqs | cardinality reduction | F1 vs. prior methods | fewer LLM calls vs. SOTA |

**Takeaway 1:** As hypothesized, log sequences are permutations of atomic and recurring patterns (lower-level KRONE-Seqs).

**Takeaway 2:** The reusability of atomic patterns enables scalable LLM detection and knowledge-driven log analysis — LLM invoked on just 1.1%–3.3% of test data

# Conclusion

Key contributions of KRONE

**01**

### Log Abstraction

Proposes an innovative log abstraction model

**02**

### Structure Recovery

Recovers meaningful structure in flat log files

**03**

### Hierarchical Detection

Offers hierarchical log anomaly detection

**04**

### Optimization

Integrates effective optimization strategies

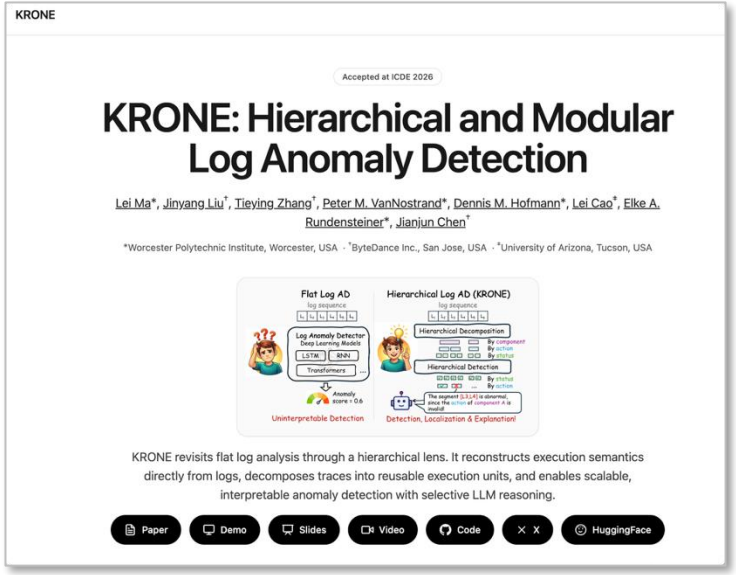
**05**

### Continuous Improvement

Continuously improves detection performance

*KRONE — innovative, structured, hierarchical, optimized, and ever-improving*

# More Information about KRONE




**Visit our webpage for more details!**

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